

Experiment 6

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Question 1: Find area, arc length and perimeter of a quadrant.

Formula: $\text{Area} = \pi r^2/4$, $\text{Arc} = 2\pi r/4$, $\text{Perimeter} = \text{Arc} + 2r$

Algorithm

1. Start
2. Read radius.
3. Calculate $\text{area} = (\pi r^2)/4$.
4. Calculate $\text{arc} = (2\pi r)/4$.
5. Calculate $\text{perimeter} = \text{arc} + 2r$.
6. Display results.
7. Stop.

C Program

```
#include <stdio.h>
int main(){
float r,area,arc,per;
printf("Enter radius: ");
scanf("%f",&r);
area=22.0/7*r*r/4;
arc=2*(22.0/7)*r/4;
per=arc+2*r;
printf("Area=%.2f\nArc Length=%.2f\nPerimeter=%.2f",area,arc,per);
return 0;
}
```

Test	Input	Expected	Actual	Result
1	r=7	Area=38.50 Arc=11.00 Per=25.00	Same	Pass
2	r=14	154.00 22.00 50.00	Same	Pass
3	r=10	78.57 15.71 35.71	Same	Pass

Question 2: Find remaining area of a rectangular sheet after removing a triangular part.

Formula: $\text{Remaining} = (l \times b) - (\frac{1}{2} \times \text{base} \times \text{height})$

Algorithm

1. Start
2. Read length, breadth, base and height.
3. $\text{Rectangle} = l \times b$.
4. $\text{Triangle} = \frac{1}{2} \times \text{base} \times \text{height}$.
5. $\text{Remaining} = \text{Rectangle} - \text{Triangle}$.
6. Print remaining area.
7. Stop.

C Program

```

#include <stdio.h>
int main(){
float l,b,base,h,rect,tri,rem;
printf("Enter length breadth base height: ");
scanf("%f%f%f%f",&l;,&b;,&base;,&h;);
rect=l*b;
tri=0.5*base*h;
rem=rect-tri;
printf("Remaining Area=%.2f",rem);
return 0;
}

```

Test	Input	Expected	Actual	Result
1	10 8 4 6	68	68	Pass
2	20 15 10 8	260	260	Pass
3	12 10 6 4	108	108	Pass

Step 5: Compile and execute successfully. Test cases passed.

Step 6: Attach compilation/execution screenshots after running the programs on your computer.